

Lab Title: Case Study 1 - iOS 13.4.1 Evidence Investigation

Student Name: Shina Ibrahim

Student ID: 2026/DFIT/17036

Course Name: Mobile and IoT Forensics (CIP-B106)

Date of Submission: 9/26/2026

Instruction Name: Aminu Idris

Platform Used: Kali Linux

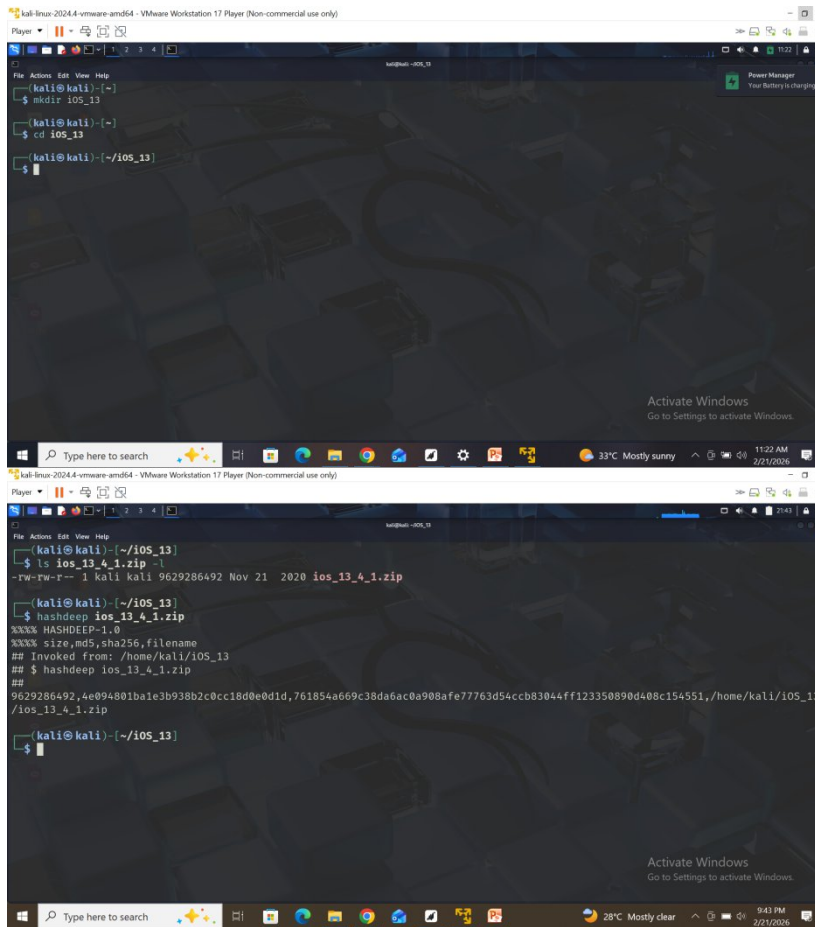
Executive Summary

This report examines the supplied iOS 13.4.1 forensic image to identify device configuration, account information, communications, application, web, media, and location-related activity. Relevant artefacts were examined using approved forensic tools and manually verified where required.

The findings were correlated across multiple sources to establish a supported chronology within a selected 24–72-hour period. Conclusions are limited to what the available evidence can demonstrate. Missing, incomplete, or altered artefacts, timestamp uncertainty, and the limitations of individual sources are considered throughout the analysis.

iOS 13 Image

1. Downloading iOS 13 Image and verify hashes



```
(kali@kali)-[~]
$ mkdir ios_13

(kali@kali)-[~]
$ cd ios_13

(kali@kali)-[~/ios_13]
$ ls ios_13_4_1.zip -l
-rw-rw-r-- 1 kali kali 9629286492 Nov 21 2020 ios_13_4_1.zip

(kali@kali)-[~/ios_13]
$ hashdeep ios_13_4_1.zip
XXXX size,md5,sha256,filename
## Invoked from: /home/kali/ios_13
## $ hashdeep ios_13_4_1.zip
##
9629286492,4e094801ba1e3b938b2c0cc18d0e0d1d,761854a669c38da6acc0a908afe77763d54ccb83044ff123350890d408c154551,/home/kali/ios_13/ios_13_4_1.zip

(kali@kali)-[~/ios_13]
$
```

```
Player | 1 2 3 4 |
kali-linux-2024.4-vmware-amd64 - VMware Workstation 17 Player (Non-commercial use only)
File Actions Edit View Help
$ tree -f Pixel_3/data/system/usagstats/0' -L 2
Pixel_3/data/system/usagstats/0
├── daily
│   ├── 1580917965671
│   ├── 1581004373881
│   ├── 1581090879070
│   ├── 1581177279272
│   ├── 1581263873664
│   ├── 1581350284755
│   ├── 1581436707859
│   ├── 1581523178051
│   ├── 1581609587889
│   └── 1581696001773
├── monthly
│   └── 1580312256162
├── version
├── weekly
│   ├── 1580312256162
│   ├── 1580917965671
│   └── 1581523178051
└── yearly
    └── 1580312256162

5 directories, 16 files

(kali@kali)~[/pixel3]
Activate Windows
Go to Settings to activate Windows.
```

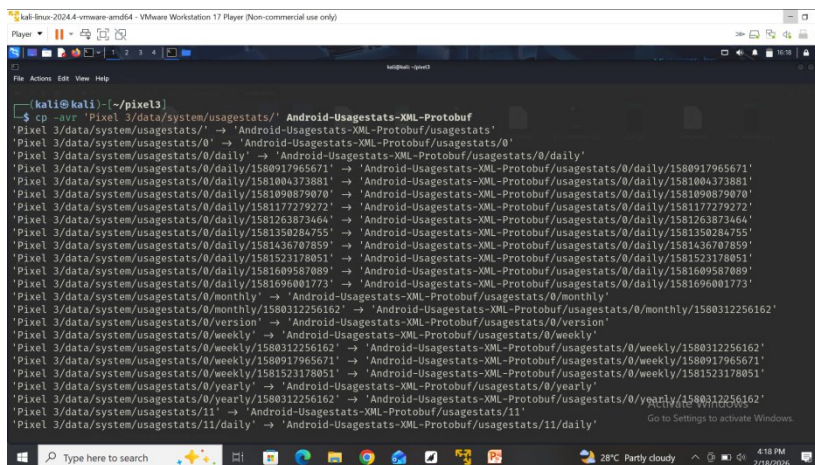
2.

```
Player | 1 2 3 4 |
kali-linux-2024.4-vmware-amd64 - VMware Workstation 17 Player (Non-commercial use only)
File Actions Edit View Help
(kali@kali)~[/pixel3]
$ date -d '@1580917965671' -u
Mon Apr 18 03:21:11 PM UTC 52067

(kali@kali)~[/pixel3]
Activate Windows
Go to Settings to activate Windows.
```

```
Player | 1 2 3 4 |
kali-linux-2024.4-vmware-amd64 - VMware Workstation 17 Player (Non-commercial use only)
File Actions Edit View Help
(kali@kali)~[/pixel3]
$ git clone https://github.com/abrignoni/Android-Usagstats-XML-Protobuf.git
Cloning into 'Android-Usagstats-XML-Protobuf' ...
remote: Enumerating objects: 83, done.
remote: Total 83 (delta 0), reused 0 (delta 0), pack-reused 83 (from 1)
Receiving objects: 100% (83/83), 90.31 KiB | 9.00 KiB/s, done.
Resolving deltas: 100% (37/37), done.

(kali@kali)~[/pixel3]
Activate Windows
Go to Settings to activate Windows.
```



```
(kali@kali) ~/pixel3
$ cp -avr 'Pixel 3/data/system/usagestats/' 'Android-UsageStats-XML-Protobuf'
'Pixel 3/data/system/usagestats/' → 'Android-UsageStats-XML-Protobuf/usagestats'
'Pixel 3/data/system/usagestats/0' → 'Android-UsageStats-XML-Protobuf/usagestats/0'
'Pixel 3/data/system/usagestats/0/daily' → 'Android-UsageStats-XML-Protobuf/usagestats/0/daily'
'Pixel 3/data/system/usagestats/0/daily/1580917965671' → 'Android-UsageStats-XML-Protobuf/usagestats/0/daily/1580917965671'
'Pixel 3/data/system/usagestats/0/daily/1581004373881' → 'Android-UsageStats-XML-Protobuf/usagestats/0/daily/1581004373881'
'Pixel 3/data/system/usagestats/0/daily/1581090879070' → 'Android-UsageStats-XML-Protobuf/usagestats/0/daily/1581090879070'
'Pixel 3/data/system/usagestats/0/daily/1581177279272' → 'Android-UsageStats-XML-Protobuf/usagestats/0/daily/1581177279272'
'Pixel 3/data/system/usagestats/0/daily/1581263873464' → 'Android-UsageStats-XML-Protobuf/usagestats/0/daily/1581263873464'
'Pixel 3/data/system/usagestats/0/daily/1581350284755' → 'Android-UsageStats-XML-Protobuf/usagestats/0/daily/1581350284755'
'Pixel 3/data/system/usagestats/0/daily/1581436707859' → 'Android-UsageStats-XML-Protobuf/usagestats/0/daily/1581436707859'
'Pixel 3/data/system/usagestats/0/daily/1581523178051' → 'Android-UsageStats-XML-Protobuf/usagestats/0/daily/1581523178051'
'Pixel 3/data/system/usagestats/0/daily/1581609587089' → 'Android-UsageStats-XML-Protobuf/usagestats/0/daily/1581609587089'
'Pixel 3/data/system/usagestats/0/daily/1581696001773' → 'Android-UsageStats-XML-Protobuf/usagestats/0/daily/1581696001773'
'Pixel 3/data/system/usagestats/0/monthly' → 'Android-UsageStats-XML-Protobuf/usagestats/0/monthly'
'Pixel 3/data/system/usagestats/0/monthly/1580312256162' → 'Android-UsageStats-XML-Protobuf/usagestats/0/monthly/1580312256162'
'Pixel 3/data/system/usagestats/0/version' → 'Android-UsageStats-XML-Protobuf/usagestats/0/version'
'Pixel 3/data/system/usagestats/0/weekly' → 'Android-UsageStats-XML-Protobuf/usagestats/0/weekly'
'Pixel 3/data/system/usagestats/0/weekly/1580312256162' → 'Android-UsageStats-XML-Protobuf/usagestats/0/weekly/1580312256162'
'Pixel 3/data/system/usagestats/0/weekly/1580917965671' → 'Android-UsageStats-XML-Protobuf/usagestats/0/weekly/1580917965671'
'Pixel 3/data/system/usagestats/0/weekly/1581523178051' → 'Android-UsageStats-XML-Protobuf/usagestats/0/weekly/1581523178051'
'Pixel 3/data/system/usagestats/0/yearly' → 'Android-UsageStats-XML-Protobuf/usagestats/0/yearly'
'Pixel 3/data/system/usagestats/0/yearly/1580312256162' → 'Android-UsageStats-XML-Protobuf/usagestats/0/yearly/1580312256162'
'Pixel 3/data/system/usagestats/11' → 'Android-UsageStats-XML-Protobuf/usagestats/11'
'Pixel 3/data/system/usagestats/11/daily' → 'Android-UsageStats-XML-Protobuf/usagestats/11/daily'
```

Conclusion

The examination of the supplied iOS 13.4.1 forensic image identified relevant evidence relating to the device configuration, communications, application and web activity, media, and system activity. Correlation of multiple artefact sources allowed a supported chronology of activity to be established for the selected examination period.

The findings are limited to the information preserved in the supplied evidence. The artefacts do not, by themselves, establish who physically operated the device or prove a person's physical

presence at a particular location. Any missing, incomplete, or uncertain data has been identified and considered when interpreting the results.